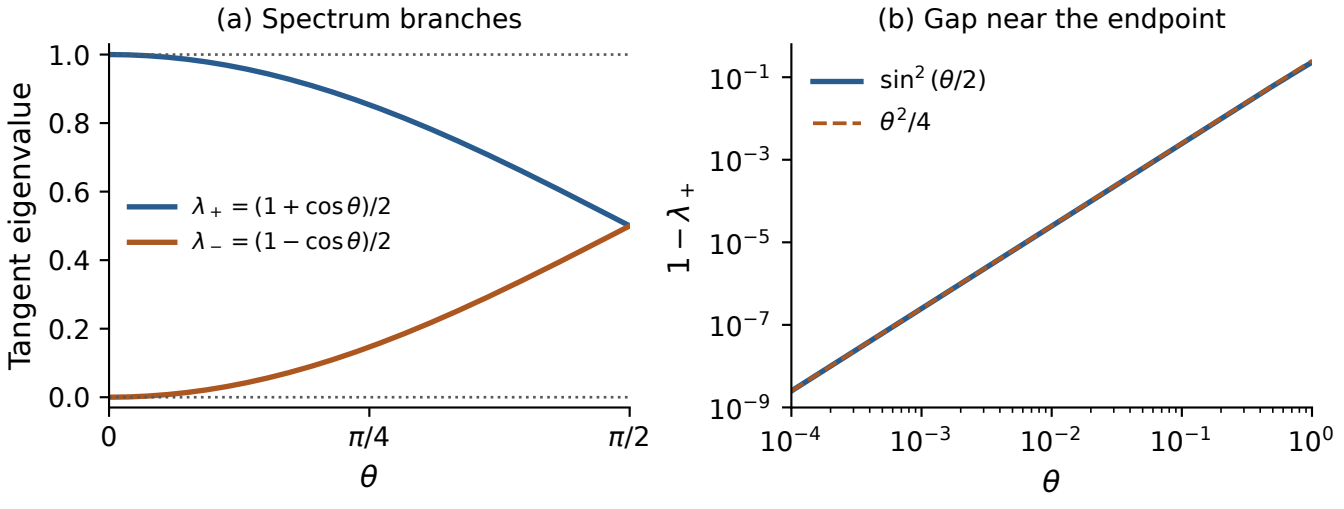
(a) Spectrum branches
Tangent eigenvalue
1.0
0.8
0.6
0.4
0.2
0.0
$\lambda_+ = (1+\cos\theta)/2$
$\lambda_- = (1-\cos\theta)/2$
0
$\pi/4$
$\pi/2$
θ
(b) Gap near the endpoint
$1-\lambda_+$
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
$\sin^2(\theta/2)$
$\theta^2/4$
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
θ